Received: 12 November 2021 | Revised: 26 May 2022 | Accepted: 11 June 2022

DOI: 10.1111/ceo.14126

ORIGINAL ARTICLE - CLINICAL SCIENCE

Clinical & Experimental Ophthalmology

WILEY

Evaluation of artificial intelligence models for the detection of asymmetric keratoconus eyes using Scheimpflug tomography

Zhe Xu MD, PhD[1] | Ruiwei Feng BE[2] | Xiuming Jin MD, PhD[1] | Heping Hu MSc[2] | Shuang Ni MD, PhD[1] | Wen Xu MD, PhD[1] | Xiangshang Zheng MD[2] | Jian Wu PhD[3,1] | Ke Yao MD[1]

[1]Eye Center of the Second Affiliated Hospital, School of Medicine, Zhejiang University, Hangzhou, Zhejiang, China

[2]College of Computer Science and Technology, Zhejiang University, Hangzhou, Zhejiang, China

[3]School of Public Health, Zhejiang University, Hangzhou, Zhejiang, China

Correspondence

Ke Yao and Jian Wu, Eye Center of the Second Affiliated Hospital, School of Medicine, Zhejiang University, No. 88 Jiefang Road, Hangzhou, Zhejiang, China, 310009

Email: xlren@zju.edu.cn and wujian2000@zju.edu.cn

Funding information

Youth Program of National Nature Science Foundation of China, Grant/Award Numbers: 81800877, 81800809; Zhejiang Province Key Research and Development Program, Grant/Award Number: 2020C03035; National Key Research and Development Program of China, Grant/Award Numbers: 2020YFE0204400, 2019YFB1404802; Zhejiang Public Welfare Technology Research Project, Grant/Award Number: LGF20F020013; Leading Innovative and Entrepreneur Team Introduction Program of Zhejiang, Grant/Award Number: 2019R01007

Abstract

Background: To evaluate artificial intelligence (AI) models based on objective indices and raw corneal data from the Scheimpflug Pentacam HR system (OCULUS Optikgeräte GmbH, Wetzlar, Germany) for the detection of clinically unaffected eyes in patients with asymmetric keratoconus (AKC) eyes.

Methods: A total of 1108 eyes of 1108 patients were enrolled, including 430 eyes from normal control subjects, 231 clinically unaffected eyes from patients with AKC, and 447 eyes from keratoconus (KC) patients. Eyes were divided into a training set (664 eyes), a test set (222 eyes) and a validation set (222 eyes). AI models were built based on objective indices (XGBoost, LGBM, LR and RF) and entire corneal raw data (KerNet). The discriminating performances of the AI models were evaluated by accuracy and the area under the ROC curve (AUC).

Results: The KerNet model showed great overall discriminating power in the test (accuracy = 94.67%, AUC = 0.985) and validation (accuracy = 94.12%, AUC = 0.990) sets, which were higher than the index-derived AI models (accuracy = 84.02%–86.98%, AUC = 0.944–0.968). In the test set, the KerNet model demonstrated good diagnostic power for the AKC group (accuracy = 95.24%, AUC = 0.984). The validation set also proved that the KerNet model was useful for AKC group diagnosis (accuracy = 94.12%, AUC = 0.983).

Conclusions: KerNet outperformed all the index-derived AI models. Based on the raw data of the entire cornea, KerNet was helpful for distinguishing clinically unaffected eyes in patients with AKC from normal eyes.

KEYWORDS

asymmetric keratoconus, deep learning, Scheimpflug tomography

Zhe Xu and Ruiwei Feng contributed equally as co-first authors.

1 | INTRODUCTION

Detecting the earliest manifestations of keratoconus (KC) is critical during preoperative screening for corneal refractive surgery. Misdiagnosis of early KC with good visual acuity and no specific corneal findings will lead to an increased risk of iatrogenic ectasia after refractive surgery.[1,2] Since KC is a progressive disease, a variety of devices and indices have been studied to identify the earliest stage of corneal ectatic disorder.[3–5] However, early detection of KC remains challenging, owing to subtle alterations in the corneal structure.

The exact definition of early KC is ambiguous. According to previous studies, there were several terms that describe the earliest stage of KC, such as asymmetric keratoconus (AKC), subclinical KC, forme fruste keratoconus (FFKC) and KC suspect.[4,6] Based on corneal pachymetry, epithelial thickness profile and both anterior and posterior curvature and elevation, individual objective indices have shown potential for early KC detection. Additionally, combining indices from different imaging methods were reported to distinguish asymmetric/subclinical KC from normal populations, with varying degrees of success.[3,5] Hwang et al.[3] reported that combining indices from the Scheimpflug imaging system and spectral domain OCT system using multivariable logistic regression analysis were useful to distinguish unaffected eyes in patients with AKC. Luz et al.[5] studied a combination of parameters with the Scheimpflug imaging system and biomechanics system, demonstrating an ability to differentiate FFKC. However, the entire cornea contains 3D spatial gradient information. The processed indices may lose subtle characteristic changes in the earliest KC stage, incurring a significant loss of data. The value of different indices for identifying the earliest manifestation of KC eyes thus remains controversial.

Using Scheimpflug imaging technology, the Pentacam HR system (OCULUS Optikgeräte GmbH, Wetzlar, Germany) has made it possible to capture the raw curvature, elevation and pachymetry data of the entire cornea within a 10-mm diameter region.[3,5,7] Artificial intelligence (AI) models have been proven to be efficient in early KC diagnosis with large data sets for input training.[8,9] Xie et al.[7] developed an AI model based on a deep learning technique called the Pentacam InceptionResNetV2 Screening System (PIRSS), which processed corneal tomographic images for KC suspect and early KC identification. This PIRSS deep learning model improved the overall screening sensitivity and specificity for early KC detection with the training of multilayer artificial neural networks. However, there remains an information gap between raw corneal spatial gradient data and tomographic images. An AI model trained with primitive and complete 3D raw data may further improve the performance of early KC detection.

In this study, we developed AI models based on objective indices and entire corneal tomographic data using the Pentacam HR system to identify the clinically unaffected eyes in patients with AKC, KC and normal eyes. The AI models were then compared with evaluate the diagnostic performances of early KC detection.

2 | METHODS

2.1 | Subjects

This retrospective study was approved by the Ethical Committee of the Second Affiliated Hospital of Zhejiang University School of Medicine. The tenets of the Declaration of Helsinki were followed for all research procedures. Written consent form was obtained from each subject. Complete ocular examinations were performed, including a review of medical and family history, best corrected visual acuity (BCVA), slit-lamp biomicroscopy, retinoscopy, ophthalmoscopy and corneal tomography using Pentacam HR system (OCULUS Optikgeräte GmbH). All Both eyes in all patients were evaluated using the Pentacam HR system. Only one eye of KC and normal patients was randomly included using Microsoft Excel (Microsoft Crop) random number generator. Patients were classified into the KC, AKC or normal groups as follows:

1. KC eyes were diagnosed using the following criteria established by the Collaborative Longitudinal Evaluation of Keratoconus (CLEK) study[3,10,11]: (1) at least one of the following slit-lamp signs: focal stromal thinning, Vogt's strias, Fleischer's ring >2-mm arc, or corneal scarring consistent with KC; (2) irregular cornea with focal steeping determined via distorted keratometry test, and distortion of the retinoscopic or ophthalmoscopic red reflex and (3) no history of contact lens wear, ocular surgery or extensive scarring.
2. AKC eyes were defined as the unaffected and asymptomatic fellow eye of each patient with unilateral KC,[3,8,10,12] according to the following criteria: (1) no clinical signs of KC on slit-lamp biomicroscopy, retinoscopy, and ophthalmoscopy; (2) no definitive abnormalities on topographic patterns for suspicious KC and KC; (3) BCVA ≥1.0 and (4) no history of contact lens wear, ocular surgery or trauma.

3. Normal eyes were enrolled among patients undergoing a routine ophthalmological examination, who met the following screening criteria: (1) normal clinical evaluations, defined as no clinical signs or suggestive topographic patterns for suspicious KC, KC or pellucid marginal degeneration; (2) no family history of KC; (3) no history of ocular surgery or trauma and (4) stopped contact lens wear for ≥3 weeks for rigid gas permeable, and ≥1 week for soft contact lenses.

2.2 | Pentacam HR system acquisition

The Pentacam HR system is a Scheimpflug-based tomography system with a rotating camera, able to record 500 measurement points from a total of 50 single-slit images. A total of 25 000 measurement points were used to assess the anterior and posterior interfaces of the cornea, containing the 3D geometric and structural information of the entire cornea. All procedures were performed by an experienced operator, and all subjects were asked to blink once before image acquisition. Patients were instructed to keep their eyes open for ~2 s while scanning. Only when "Examination Quality Specification" indicated 'OK', were the scanning results accepted. Objective indices from the built-in Pentacam HR software (version 6.02r23) were obtained as follows:

AK1: flat keratometry of anterior cornea.

AK2: steep keratometry of anterior cornea.

AKm: average keratometry of AK1 and AK2.

PK1: steep keratometry of posterior cornea.

PK2: flat keratometry of posterior cornea.

PKm: average keratometry of PK1 and PK2.

ISV: index of surface variance;

IVA: index of vertical asymmetry.

KI: keratoconus index.

CKI: central keratoconus index.

IHA: index of height asymmetry.

IHD: index of height decentration.

Rmin: minimum sagittal curvature.

TCT: the thinnest cornea thickness.

IT: distance from corneal apex to thinnest location.

AE: elevation at the corneal apex of the anterior surface over the 8-mm best-fit sphere (BFS) float mode.

PE: elevation at the corneal apex of the anterior surface over the 8-mm BFS float mode.

PPImin: minimum pachymetric progression index.

PPImax: maximum pachymetric progression index.

PPIavg: average pachymetric progression index.

ARTmax: maximum Ambrósio relational thickness.

Df: deviation of front surface elevation difference.

Db: deviation of back surface elevation difference.

Dp: deviation of average pachymetric progression.

Dt: deviation of minimum thickness.

Da: deviation of ARTmax.

BAD-D: Belin- Ambrósio Deviation scores.

The built-in Pentacam HR software was used to obtain the raw data of five matrices, including the anterior and posterior corneal curvatures, anterior and posterior elevations, and entire pachymetry distribution. The U12 files containing the raw data of the five matrices within a 10-mm diameter region were exported, transferred, and tabulated in CSV files.

2.3 | AI models

Based on the AI models, subjects were classified into one of three categories: normal, AKC and KC. Five AI models were compared, including four machine-learning algorithms, and a proposed convolutional neural networks-based method called KerNet.[8] These four machine learning algorithms used the objective indices obtained as the input data, including the eXtreme Gradient Boosting (XGBoost),[13] Light Gradient Boosting Machine (LGBM),[14] Logistic Regression (LR),[15] and Random Forest (RF) algorithms.[16] The state-of-the-art performances for the classification tasks of each algorithm were evaluated.[17] More precisely, we trained the models on the training set, and optimised model parameters based on their performances on the test set. Finally, we evaluated the models using the external validation set, and presented the model performances using both the test and external validation sets. Briefly, XGBoost and LGMB base on the Gradient Boosting Decision Tree machine learning algorithms. XGBoost firstly performs a pre-sort step to rank each feature, subsequently updates tree nodes using a depth-wise strategy.[13] LGMB uses a histogram-based decision tree, and updates tree nodes using a leaf-wise growth strategy.[14] The LR algorithm introduces a logistic function to generate the model, and is optimised with specific loss functions.[15] RF is a decision tree-based algorithm.[16] By assembling more than one decision tree for classification, the RF can outperform many machine-learning algorithms, and run efficiently on large data set.

KerNet is a prudently designed, end-to-end, deep learning network for early KC and KC detection.[8] The algorithm takes utilises the entire raw 3D data set of the five matrices—derived from the Pentacam HR system—as the input, analysing the raw data within the 10-mm diameter region. KerNet consists of five branches, each of which takes one slice as the input. To fuse the extracted features of different slices by moderately considering the potential relationships among slices, the ideas of multilevel fusion and spatial attention mechanism are introduced into this novel structure (Figure 1). Previous study has reported the

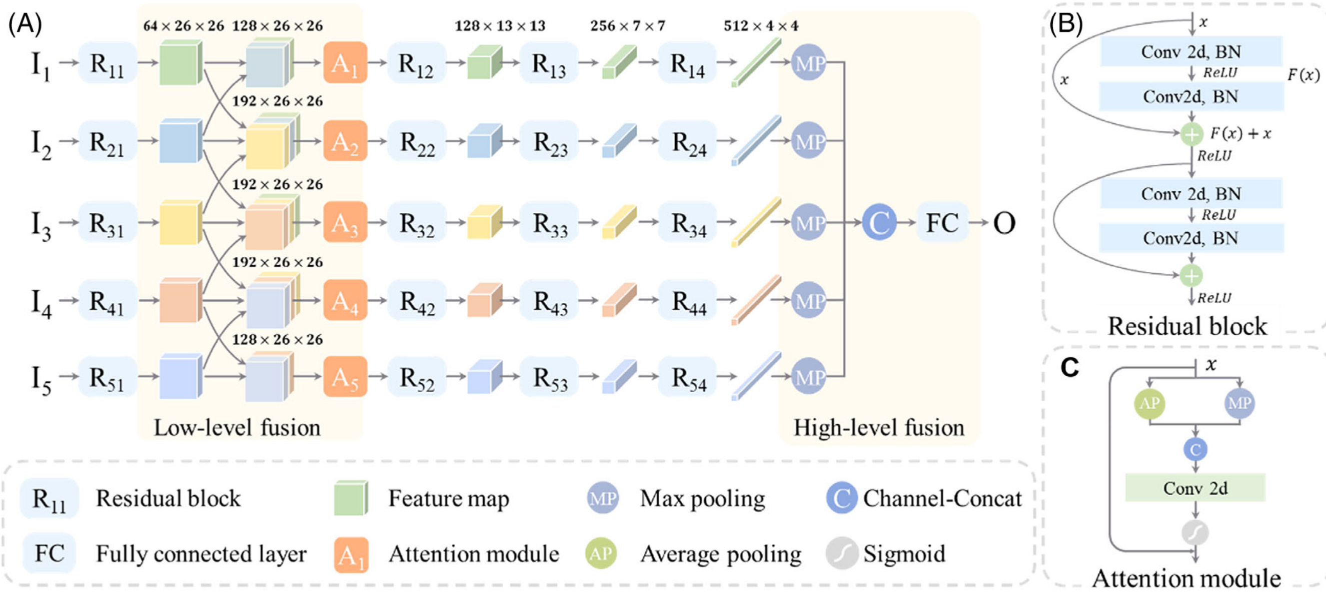

FIGURE 1 Illustration of the KerNet model. (A) Overall structure of the KerNet model; (B) Detailed structure of Residual block; (C) Detailed structure of Attention module

details of the KerNet model.[8] Moreover, the Gradient-weighted Class Activation Mapping (Grad-CAM) is generated to visualise the average attention for diagnostic regions (Figure 2).[18] The red regions correspond with high scores for AKC and KC detection, while the blue regions represent low scores and low attention.

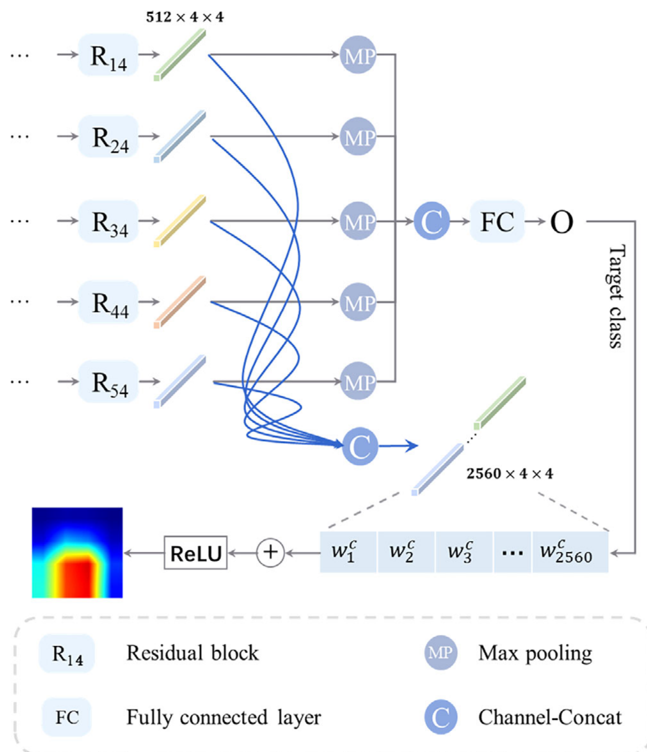

FIGURE 2 Illustration of specific Grad-CAM operation for average attention maps

2.4 | Statistical analysis

The means and *SD* were calculated for the demographic data and objective machine-derived indices. Data analyses were performed using the Statistical Package for the Social Sciences software (SPSS ver. 17; IBM). Comparisons among the normal, AKC and KC groups were performed using one-way analysis of variance (ANOVA). Statistical significance was defined as $p < 0.05$.

Receiver operating characteristic (ROC) curves for deep learning models of the three categories were calculated to obtain the area under the ROC curves (AUCs). Given the true positives (TP), false positives (FP), true negatives (TN) and false negatives (FN) of the three categories, the metrics were calculated to measure the classification performances of all the deep learning models as follows:

$$\text{Sensitivity} = \text{TP}/(\text{TP}+\text{FN})$$

$$\text{Specifity} = \text{TN}/(\text{FP}+\text{TN})$$

$$\text{Precision} = \text{TP}/(\text{TP}+\text{FP})$$

$$\text{Accuracy} = (\text{TP}+\text{TN})/(\text{TP}+\text{FP}+\text{TN}+\text{FN})$$

$$\text{F1} = 2 \cdot \text{Sensitivity} \cdot \text{Precision}/(\text{Sensitivity}+\text{Precision})$$

TABLE 1 Objective indices of all subjects for normal, AKC and KC groups

Indices	Normal	AKC	KC	*p*-value AKC versus normal	KC versus normal	AKC versus KC
AK1 (D)	42.4 ± 2.5	42.5 ± 3.1	48.4 ± 11.6	0.712	<0.001	<0.001
AK2 (D)	43.6 ± 2.0	43.9 ± 1.7	50.5 ± 6.8	0.058	<0.001	<0.001
AKm (D)	42.9 ± 3.0	43.3 ± 1.6	49.1 ± 6.1	0.064	<0.001	<0.001
PK1 (D)	6.1 ± 0.2	6.1 ± 0.3	7.1 ± 1.1	0.823	<0.001	<0.001
PK2 (D)	6.5 ± 0.3	6.5 ± 0.4	7.7 ± 1.2	0.480	<0.001	<0.001
PKm (D)	6.3 ± 0.3	6.3 ± 0.3	7.5 ± 1.5	0.927	<0.001	<0.001
ISV	19.19 ± 12.65	29.05 ± 20.88	90.70 ± 42.86	<0.001	<0.001	<0.001
IVA	0.12 ± 0.07	0.26 ± 0.22	0.84 ± 0.43	<0.001	<0.001	<0.001
KI	1.02 ± 0.02	1.06 ± 0.05	1.22 ± 0.15	<0.001	<0.001	<0.001
CKI	1.01 ± 0.15	1.01 ± 0.02	1.11 ± 0.33	0.929	<0.001	<0.001
IHA	6.04 ± 6.73	11.51 ± 15.47	28.76 ± 23.49	<0.001	<0.001	<0.001
IHD	0.01 ± 0.05	0.03 ± 0.03	0.13 ± 0.08	<0.001	<0.001	<0.001
Rmin	7.59 ± 0.29	7.43 ± 0.39	5.94 ± 0.84	<0.001	<0.001	<0.001
TCT (μm)	526.87 ± 44.08	511.00 ± 32.52	453.72 ± 51.51	<0.001	<0.001	<0.001
IT (mm)	0.64 ± 0.24	0.70 ± 0.24	0.79 ± 0.42	<0.05	<0.001	<0.001
AE (μm)	1.77 ± 2.55	3.88 ± 2.82	23.24 ± 14.77	<0.001	<0.001	<0.001
PE (μm)	4.09 ± 3.20	9.83 ± 7.18	51.86 ± 30.45	<0.001	<0.001	<0.001
PPImin	0.84 ± 0.35	0.82 ± 0.23	1.48 ± 0.81	0.565	<0.001	<0.001
PPImax	1.43 ± 0.44	1.67 ± 0.38	3.14 ± 1.70	<0.001	<0.001	<0.001
PPIavg	1.21 ± 0.40	1.29 ± 0.29	2.12 ± 0.90	<0.05	<0.001	<0.001
ARTmax	391.92 ± 94.70	339.85 ± 87.84	151.65 ± 52.17	<0.001	<0.001	<0.001
Df	0.84 ± 1.94	1.35 ± 1.45	13.19 ± 9.14	<0.05	<0.001	<0.001
Db	−0.21 ± 3.83	0.73 ± 1.02	8.97 ± 7.13	<0.05	<0.001	<0.001
Dp	1.64 ± 2.54	1.84 ± 1.51	9.71 ± 6.73	0.296	<0.001	<0.001
Dt	0.36 ± 1.20	0.70 ± 0.94	2.83 ± 1.65	<0.001	<0.001	<0.001
Da	0.81 ± 0.86	1.25 ± 0.80	3.07 ± 0.73	<0.001	<0.001	<0.001
BAD-D	0.98 ± 0.62	1.84 ± 0.94	9.56 ± 5.32	<0.001	<0.001	<0.001

Note: Data are shown as mean and *SD*.
Abbreviations: AKC, asymmetric keratoconus group; KC, keratoconus group; Normal, normal group.

3 | RESULTS

3.1 | Demographics

All total, 1108 eyes of 1108 patients were enrolled, including 430 normal subjects (114 females and 316 males; 25.0 ± 5.3 years), 231 patients with AKC (57 females and 174 males; 24.1 ± 6.2 years), and 447 patients with KC (119 females and 328 males; 25.1 ± 9.0 years). The spherical equivalent (SE, Diopter) results of the normal (−4.93 ± 2.07 D) and AKC (−4.47 ± 2.44 D) groups differed from those of the KC group (−7.07 ± 4.45 D; ANOVA, $p < 0.05$). The BCVA (decimal VA) results of the normal (1.01 ± 0.05 D) and AKC (1.01 ± 0.05 D) groups were higher than those of the KC group (0.51 ± 0.22 D; ANOVA, $p < 0.05$).

3.2 | Objective machine-derived indices

Table 1 showed the objective machine-derived indices derived from the built-in Pentacam HR system for all groups. The anterior corneal refractive power (AK1, AK2, AKm), posterior corneal refractive power (PK1, PK2, PKm), central keratoconus index (CKI), minimum pachymetric progression index (PPImin), and deviation

TABLE 2 Overall discriminating performances of objective indices-derived models and KerNet model based on entire corneal raw data

	Models	Sensitivity (%)	Specificity (%)	Precision (%)	Accuracy (%)	F1 (%)	AUC
Test Set	XGBoost	81.18	92.72	84.62	85.88	81.94	0.960
	LGBM	80.17	92.21	82.58	84.71	80.77	0.955
	LR	82.30	92.51	84.46	85.88	82.97	0.944
	RF	82.80	93.17	84.55	86.47	83.37	0.960
	KerNet	95.01	97.64	92.64	94.67	93.56	0.985
Validation Set	XGBoost	77.30	92.16	79.60	84.62	77.88	0.965
	LGBM	79.08	92.29	79.04	84.02	79.03	0.961
	LR	78.97	92.48	79.54	84.62	79.20	0.960
	RF	80.46	93.31	83.36	86.98	81.28	0.968
	KerNet	93.09	97.08	93.09	94.12	93.09	0.990

Abbreviations: LGBM, light gradient boosting machine; LR, logistic regression; XGBoost, eXtreme gradient boosting.

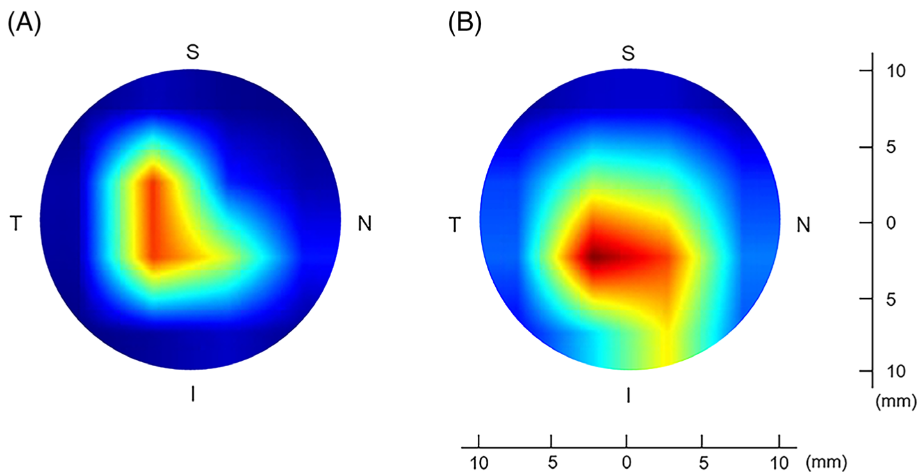

FIGURE 3 Visualisation of diagnostic attention maps generated by Grad-CAM for the AKC and KC groups. (A) Grad-CAM map for AKC group; (B) Grad-CAM map for KC group. Red region, high attention; Blue region, low attention; S, superior; T, temporal; I, inferior; N, nasal

TABLE 3 Discriminating performances of the KerNet model for normal, AKC, and KC groups

	Groups	Sensitivity (%)	Specificity (%)	Precision (%)	Accuracy (%)	F1 (%)	AUC
Test Set	Normal	92.42	98.02	96.83	95.81	94.57	0.965
	AKC	96.77	94.89	81.08	95.24	88.24	0.984
	KC	95.83	100.00	100.00	98.16	97.87	0.992
Validation Set	Normal	93.94	96.08	93.94	95.24	93.94	0.969
	AKC	86.84	96.21	86.84	94.12	86.84	0.983
	KC	98.48	98.96	98.48	98.77	98.48	1.000

Abbreviations: AKC, asymmetric keratoconus group; KC, keratoconus group; Normal, normal group.

of average pachymetric progression (Dp) showed no significant difference between the AKC and normal groups (ANOVA, $p > 0.05$). However, the rest indices were significantly different between the AKC and normal groups (Table 1; ANOVA, $p < 0.05$). All objective machine-derived indices were significantly different between the KC and normal groups (Table 1; ANOVA, $p < 0.05$).

3.3 | AI models

All the AI models were evaluated based on a training set (664 eyes), a test set (222 eyes) and an external validation set (222 eyes). Regarding the overall discriminating power of the test set, the KerNet model—based on raw corneal data—demonstrated a higher overall classification ability (accuracy = 94.67%, AUC = 0.985; Table 2) than the index-derived models, such as XGBoost (accuracy = 85.88%, AUC = 0.960; Table 2), LGBM (accuracy = 84.71%, AUC = 0.955; Table 2), LR (accuracy = 85.88%, AUC = 0.944; Table 2), and RF (accuracy = 86.47%, AUC = 0.960, Table 2). The performance on the external validation set also demonstrated the great discriminating power of the KerNet model (accuracy = 94.12% AUC = 0.990; Table 2), which was higher than that of XGBoost (accuracy = 84.62%, AUC = 0.965; Table 2), LGBM (accuracy = 84.02%, AUC = 0.961; Table 2), LR (accuracy = 84.62%, AUC = 0.960; Table 2), and RF (accuracy = 86.98%, AUC = 0.968; Table 2).

The average attention maps generated by the Grad-CAM of the KerNet model for visualisation are shown in Figure 3. They illustrate that AKC diagnosis should focus on the temporal and inferior regions with red colour (Figure 3A). Correspondingly, KerNet mainly focused on the inferior region of the KC group (Figure 3B). Using the test set, the KerNet model demonstrated good diagnostic power for AKC discrimination (accuracy = 95.24%, AUC = 0.984; Table 3). While the validation set also proved that the KerNet model was useful for AKC diagnosis (accuracy = 94.12%, AUC = 0.983; Table 3).

4 | DISCUSSION

In this study, the deep learning-derived classifier (KerNet) demonstrated great diagnostic power for identifying KC, AKC, and normal eyes. The KerNet model—based on the raw entire corneal data set—largely outperformed the objective index-derived models. Meanwhile, the KerNet model showed high accuracy regarding AKC discrimination. To date, this is the first study to generate AI models from complete 3D spatial gradient information using raw entire corneal tomographic data for early KC detection.

There has been great interest regarding the discrimination of at-risk corneas at the early KC stage. An important clinical task is to recognise patients with early KC and thereby avoid refractive surgery. Moreover, early KC patients should undergo long-term observation and accept cross-linking therapy at appropriate time to stop KC progression.[19] However, the inclusion criteria and descriptions for early KC remain controversial. Arbelaez et al.[10] considered the subclinical KC group as the early KC group, which included eyes with FFKC (the fellow eye of clinically unilateral KC) and KC suspects (eyes with subtle signs of KC but without evidence of clinical KC in opposite eye). Saad et al.[20] included contralateral eyes with FFKC in patients with KC as the early KC group. Hwang et al.[3] used the clinically unaffected eye of patients with AKC as the early KC eyes. As study reviewed, the most common subclinical KC definition was an eye with topographic signs of keratoconus and/or suspicious topographic findings under normal slit-lamp examination and keratoconus in the fellow eye. The most common form fruste keratoconus definition was an eye with normal topography, normal slit-lamp examination and keratoconus in the fellow eye.[21] It was meaningful to enrol the unaffected fellow eye of patients with no definitive abnormalities on topographic patterns for suspicious KC and KC as the early KC group in this current study. Since both eyes are genetically identical, the less-affected eye may undergo incomplete gene expression.[22,23] Additionally, previous studies have shown that the frequency of true unilateral KC is very low. Unilateral KC will develop in the opposite eye if the observation time is long enough.[24,25] Remarkably, true unilateral keratoconus does not exist.[26] Thus, it is asserted that unaffected eyes with AKC exhibiting no slit lamp findings and no definitive abnormalities on topographic patterns are preferable samples for the early KC stage.[4,6,21,27]

With the advent of Pentacam HR system, diagnosis of KC and early KC achieved due to objective indices obtained from pachymetric distribution, surface elevation and displacement of location of thinnest point. The 27 objective indices (Table 1) obtain from Pentacam HR system of each group in this current study were similar with previous study.[28,29] However, the cut off values of individual indices for KC and early KC prevented generalisation according to different studies, and should not be the only parameter.[21,27] The boundary of early KC was controversial. Thus, previous study built the Pentacam Random Forest Index (PRFI) based on AI method using these powerful indices to distinguish patients with normal topography (VAE-NT) in one eye and clinically diagnosed ectasia in the other (VAE-E).[30,31] Predictive models of combined parameters were reported, such as BAD-D index and the ABCD Grading System.[21,27,32] In the field of early KC detection, the cut off points are still improved as the research progresses. It is more rigorous to test the diagnostic parameters and models if diagnostic criteria based on the Pentacam HR system would be used in the future study.

The use of AI is an efficient technique to combine appropriate indices and functions using large amounts of training data.[33,34] With significant potential for the advancement of imaging devices and techniques, early KC screening represents a burgeoning field of study. As indices for early KC detection, combination models using the AI method were built to improve diagnostic accuracy. Arbelaez et al.[10] applied a support vector machine (SVM) to diagnose subclinical KC based on corneal measurements provided by a Scheimpflug camera combined with Placido corneal topography (Sirius; CSO, Florence, Italy).

The diagnostic accuracy of the SVM was 97.3%. Smadja et al.[35] developed an automated decision tree classifier containing 55 parameters derived from a GALILEI dual Scheimpflug analyser (Ziemer Ophthalmic Systems AG, Port, Switzerland). The discriminating power for FFKC reached 93.6% sensitivity and 97.2% specificity. Trained on indices obtained from the Pentacam system for early KC discrimination, Kovács et al.[36] built a neural network classifier (AUC = 0.96), Ruiz Hidalgo et al.[37] built a SVM classifier (accuracy = 93.1%), and Lopes et al.[30] built a random forest index (AUC = 0.992). In this study, the index-derived machine learning models (XGBoost, LGBM, LR, and RF) demonstrated overall high accuracy and AUCs for both the test (accuracy >84.71%, AUC >0.944) and validation (accuracy >84.02%, AUC >0.960) sets. All these index-based AI diagnostic models were proven to be effective for early KC discrimination. Since the study samples, sample size and algorithms were different, direct comparisons among different models were difficult. However, these extracted indices were designed to grasp the characteristic changes of the overall cornea. Additionally, the 3D spatial gradients and distributions of the entire corneal structure might be hidden and ignored, which remain space for further improvements.

Considering the subtle changes in eyes with early KC, information regarding the entire cornea was considered important. According to previous studies, deep learning models based on colour-coded mappings showed advantages over numeric indices.[38] Kamiya et al.[39] developed a ResNet-18 model—designed to detect different KC grades—including six colour-coded maps, measured with the swept-source OCT CASIA SS-1000 (Tomey Corporation). The ResNet-18 model contained anterior elevation, anterior curvature, posterior elevation, posterior curvature, total refractive power and pachymetry maps. The mean output data demonstrated an accuracy of 0.991. Xie et al.[7] used a Pentacam InceptionResNetV2 Screening System (PIRSS) to analyse four tomographic images, including axial curvature, front elevation, back elevation and corneal thickness. The early KC discrimination using the PIRSS demonstrated 92.0% sensitivity and 99.1% specificity. However, the uninterpretable and unstructured raw data of corneal tomography may contain more relevant clinical features. Issarti et al.[9] reported a computer aided diagnosis (CAD) system concerning the anterior, posterior corneal elevation raw data for a 7-mm diameter and minimum corneal thickness. The CAD system was able to detect suspect KC with 96.56% accuracy, 97.78% sensitivity and 95.56% specificity. Our study built the KerNet model, applying five matrices corresponding with the 10-mm diameter raw data as input for early KC discrimination. The five matrices included anterior curvature, posterior curvature, anterior elevation, posterior elevation and pachymetry. The KerNet model yielded better diagnostic performance for early KC discrimination than the index-derived models, with overall accuracies of 95.24% in the test set (AUC = 0.984) and 94.12% in the validation set (AUC = 0.983) for AKC detection. To the best of our knowledge, this is the first end-to-end deep learning model based on complete entire corneal data. The KerNet model was carefully designed to capture features of the entire corneal raw data by introducing the idea of multilevel fusion, offering a new way to apply deep learning methods for early KC detection.

The Grad-CAM visualisation maps showed important regions for predicting attention. The Grad-CAM of the AKC group illustrated subtle characteristic changes have already excited in the entire tomography raw data. Moreover, it is worth noting that the early KC tomography changes tended to be located in the inferior and temporal regions. The diagnostic attention in the KC group was focused on the inferior region. In terms of previous studies, an abnormal Bowman's layer thickness profile was proven in subclinical KC eyes using the UHR-OCT system.[40] While collagen fibre changes were also captured in KC suspects using anterior segment polarisation-sensitive OCT.[41] Additionally, Kitazawa1 et al.[42] reported an imbalance of anterior and posterior corneal surfaces in the early stage of KC. All these corneal layer structural and surface changes may explain the subtle variations in 3D spatial gradient information. The Grad-CAM results of patients with AKC therefore support the idea that complete raw tomography data of the entire cornea may contain markers for early KC diagnosis.

This study has several limitations. First, without an accessible public data set for early KC research purposes, the quality and quantity of the concerned samples are limited. Unified data may thus be beneficial for optimization and comparison among different AI models. Second, we only enrolled eyes with AKC. The diagnostic power of KerNet for different KC stages was not evaluated. In the future, we may enrich the sample size and enhance model optimization to further improve the diagnostic performance of KerNet. In addition, the traditional CLEK criteria may generally include advanced subjects into KC group as slit-lamp signs happen in typical KC stage. In further study, it is helpful to improve generalisation ability of the KerNet using diagnostic criteria based on the Pentacam HR system, which are more rigorous for subject inclusion.

In summary, based on the raw data of the entire cornea obtained using the Pentacam HR system, KerNet—using a deep learning network—was helpful for identifying the clinically unaffected eyes in patients with AKC, KC and normal eyes. In this manner, KerNet outperformed all index-based AI models, and might offer a new method for early KC screening.

FUNDING INFORMATION

This study was supported by research grants from the Youth Program of National Nature Science Foundation of China (81800877 and 81800809), the Zhejiang Province Key Research and Development Program (2020C03035), the National Key Research and Development Program of China (2020YFE0204400 and 2019YFB1404802), the Zhejiang Public Welfare Technology Research Project (LGF20F020013) and the Leading Innovative and Entrepreneur Team Introduction Program of Zhejiang (2019R01007).

CONFLICT OF INTEREST

The authors declare no conflicts of interest.

CLINICAL TRIAL REGISTRATION

ChiCTR2000039070 (Chinese Clinical Trial Registry: https://www.chictr.org.cn/).

ORCID

Xiuming Jin https://orcid.org/0000-0001-7492-5143
Ke Yao https://orcid.org/0000-0002-6764-7365

REFERENCES

1. Ambrósio R Jr, Wilson SE. Complications of laser in situ keratomileusis: etiology, prevention, and treatment. *J Refract Surg*. 2001;17:350-379.
2. Randleman JB, Woodward M, Lynn MJ, Stulting RD. Risk assessment for ectasia after corneal refractive surgery. *Ophthalmology*. 2008;115:37-50.
3. Hwang ES, Perez-Straziota CE, Kim SW, Santhiago MR, Randleman JB. Distinguishing highly asymmetric keratoconus eyes using combined Scheimpflug and spectral-domain OCT analysis. *Ophthalmology*. 2018;125:1862-1871.
4. Klyce SD. Chasing the suspect: keratoconus. *Br J Ophthalmol*. 2009;93:845-847.
5. Luz A, Lopes B, Hallahan KM, et al. Enhanced combined tomography and biomechanics data for distinguishing forme fruste keratoconus. *J Refract Surg*. 2016;32:479-494.
6. Klyce SD. The future of keratoconus screening with artificial intelligence. *Ophthalmology*. 2018;125:1872-1873.
7. Xie Y, Zhao L, Yang X, et al. Screening candidates for refractive surgery with corneal tomographic-based deep learning. *JAMA Ophthalmol*. 2020;138:519-526.
8. Feng R, Xu Z, Zheng X, et al. KerNet: a novel deep learning approach for keratoconus and sub-clinical keratoconus detection based on raw data of the Pentacam system. *IEEE J Biomed Health Inform*. 2021;25:3898-3910.
9. Issarti I, Consejo A, Jiménez-García M, Hershko S, Koppen C, Rozema JJ. Computer aided diagnosis for suspect keratoconus detection. *Comput Biol Med*. 2019;109:33-42.
10. Arbelaez MC, Versaci F, Vestri G, Barboni P, Savini G. Use of a support vector machine for keratoconus and subclinical keratoconus detection by topographic and tomographic data. *Ophthalmology*. 2012;119:2231-2238.
11. Zadnik K, Barr JT, Edrington TB, et al. Baseline findings in the collaborative longitudinal evaluation of keratoconus (CLEK) study. *Invest Ophthalmol Vis Sci*. 1998;39:2537-2546.
12. Golan O, Piccinini AL, Hwang ES, et al. Distinguishing highly asymmetric keratoconus eyes using dual scheimpflug/placido analysis. *Am J Ophthalmol*. 2019;201:46-53.
13. Chen T, He T, Benesty M, Khotilovich V, Tang Y, Cho H. Xgboost: extreme gradient boosting. R package version 04-2. 2015;1:1-4.
14. Ke G, Meng Q, Finley T, et al. Lightgbm: a highly efficient gradient boosting decision tree. *Adv Neural Inf Proces Syst*. 2017;30:3146-3154.
15. Grimm LG, Yarnold PR. *Reading and Understanding Multivariate Statistics*. American Psychological Association; 1995:217-244.
16. Breiman L. Random forests. *Mach Learn*. 2001;45:5-32.
17. Podder P, Mondal MRH. Machine learning to predict COVID-19 and ICU requirement: 2020 11th International Conference on Electrical and Computer Engineering (ICECE), 2020:483–486.
18. Selvaraju RR, Cogswell M, Das A, Vedantam R, Parikh D, Batra D. Grad-cam: visual explanations from deep networks via gradient-based localization: Proceedings of the IEEE International Conference on Computer Vision, 2017:618–626.
19. Mazzotta C, Traversi C, Baiocchi S, et al. Corneal collagen cross-linking with riboflavin and ultraviolet a light for pediatric keratoconus: ten-year results. *Cornea*. 2018;37:560-566.
20. Saad A, Gatinel D. Topographic and tomographic properties of forme fruste keratoconus corneas. *Invest Ophthalmol Vis Sci*. 2010;51:5546-5555.
21. Henriquez MA, Hadid M, Izquierdo L Jr. A systematic review of subclinical keratoconus and forme fruste keratoconus. *J Refract Surg*. 2020;36:270-279.
22. Ihalainen A. Clinical and epidemiological features of keratoconus genetic and external factors in the pathogenesis of the disease. *Acta Ophthalmol Suppl*. 1986;178:1-64.
23. Rabinowitz YS, Maumenee IH, Lundergan MK, et al. Molecular genetic analysis in autosomal dominant keratoconus. *Cornea*. 1992;11:302-308.
24. Li X, Rabinowitz YS, Rasheed K, Yang H. Longitudinal study of the normal eyes in unilateral keratoconus patients. *Ophthalmology*. 2004;111:440-446.
25. Holland DR, Maeda N, Hannush SB, et al. Unilateral keratoconus. Incidence and quantitative topographic analysis. *Ophthalmology*. 1997;104:1409-1413.
26. Gomes JA, Tan D, Rapuano CJ, et al. Global consensus on keratoconus and ectatic diseases. *Cornea*. 2015;34:359-369.
27. Martínez-Abad A, Piñero DP. New perspectives on the detection and progression of keratoconus. *J Cataract Refract Surg*. 2017;43:1213-1227.
28. Huseynli S, Salgado-Borges J, Alio JL. Comparative evaluation of Scheimpflug tomography parameters between thin non-keratoconic, subclinical keratoconic, and mild keratoconic corneas. *Eur J Ophthalmol*. 2018;28:521-534.
29. Hashemi H, Beiranvand A, Yekta A, Maleki A, Yazdani N, Khabazkhoob M. Pentacam top indices for diagnosing subclinical and definite keratoconus. *J Curr Ophthalmol*. 2016;28:21-26.

30. Lopes BT, Ramos IC, Salomão MQ, et al. Enhanced tomographic assessment to detect corneal ectasia based on artificial intelligence. *Am J Ophthalmol*. 2018;195:223-232.
31. Ambrósio R Jr, Lopes BT, Faria-Correia F, et al. Integration of Scheimpflug-based corneal tomography and biomechanical assessments for enhancing ectasia detection. *J Refract Surg*. 2017;33:434-443.
32. Belin MW, Duncan JK. Keratoconus: the ABCD grading system. *Klin Monatsbl Augenheilkd*. 2016;233:701-707.
33. Tivive FH, Bouzerdoum A. Efficient training algorithms for a class of shunting inhibitory convolutional neural networks. *IEEE Trans Neural Netw*. 2005;16:541-556.
34. Matsugu M, Mori K, Mitari Y, Kaneda Y. Subject independent facial expression recognition with robust face detection using a convolutional neural network. *Neural Netw*. 2003;16:555-559.
35. Smadja D, Touboul D, Cohen A, et al. Detection of subclinical keratoconus using an automated decision tree classification. *Am J Ophthalmol*. 2013;156:237-246.e1.
36. Kovács I, Miháltz K, Kránitz K, et al. Accuracy of machine learning classifiers using bilateral data from a Scheimpflug camera for identifying eyes with preclinical signs of keratoconus. *J Cataract Refract Surg*. 2016;42:275-283.
37. Ruiz Hidalgo I, Rodriguez P, Rozema JJ, et al. Evaluation of a machine-learning classifier for keratoconus detection based on Scheimpflug tomography. *Cornea*. 2016;35:827-832.
38. Lavric A, Valentin P. KeratoDetect: keratoconus detection algorithm using convolutional neural networks. *Comput Intell Neurosci*. 2019;2019:8162567-8162569.
39. Kamiya K, Ayatsuka Y, Kato Y, et al. Keratoconus detection using deep learning of colour-coded maps with anterior segment optical coherence tomography: a diagnostic accuracy study. *BMJ Open*. 2019;9:e031313.
40. Xu Z, Jiang J, Yang C, et al. Value of corneal epithelial and Bowman's layer vertical thickness profiles generated by UHR-OCT for sub-clinical keratoconus diagnosis. *Sci Rep*. 2016;6:31550.
41. Meek KM, Tuft SJ, Huang Y, et al. Changes in collagen orientation and distribution in keratoconus corneas. *Invest Ophthalmol Vis Sci*. 2005;46:1948-1956.
42. Kitazawa K, Itoi M, Yokota I, et al. Involvement of anterior and posterior corneal surface area imbalance in the pathological change of keratoconus. *Sci Rep*. 2018;8:14993.

How to cite this article: Xu Z, Feng R, Jin X, et al. Evaluation of artificial intelligence models for the detection of asymmetric keratoconus eyes using Scheimpflug tomography. *Clin Experiment Ophthalmol*. 2022;50(7):714-723. doi:10.1111/ceo.14126